Team - 06

Project Title: Neural Kinematics and Reinforcement Learning for Collision-Free Space Manipulator Path Planning

Level 1:

Use simulated multi-axis space manipulator joint data to develop a kinematics-aware AI/ML model that predicts collision-free trajectories and computes joint angles to prevent self collision with the satellite body.

Level 2:

Implement a lightweight Deep Reinforcement Learning (DRL) policy agent optimized for low compute overhead that dynamically re-routes the manipulator path in real time when unexpected geometric obstacles or hardware position errors occur.

Dataset:

OpenRAVE / PyBullet Manipulator Kinematics & Space Robot Simulation Dataset